AIVI Career & Campus OS — Latest Technical Documentation (August 13, 2026)

> **Document ID**: LATEST_TECHNICAL_DOCUMENTATION_2026-08-13
> **Release Version**: 2.4.0-PROD
> **Legal Entity**: AIVI Intelligence Private Limited
> **Headquarters**: Gorakhpur, Uttar Pradesh, India
> **Authoritative Baseline**: Production & Staging Verified Codebase

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1. Executive Summary & Brand Overview

AIVI (AI Voice & Career Intelligence) is India's leading purpose-built **AI Career Operating System (Career OS)** and **Campus Intelligence Platform**. Owned and operated by **AIVI Intelligence Private Limited** (headquartered in Gorakhpur, Uttar Pradesh), AIVI bridges the gap between academic preparation and recruiter expectations for students, freshers, working professionals, and higher education institutions across Bharat.

Core Value Propositions

1. **For Individual Candidates (B2C — `aivilabs.com`)**:
    - **Resume Studio v2**: 4 specialized creation paths (First Resume Wizard, Quick Edits & Metrics, Targeted JD Keyword Tailoring, and Full Structural Rebuilds). Enforces single-column Taleo, Workday, and Greenhouse ATS parser safety.
    - **Job Fit Analysis Tool**: 0–100% instant match calculation, missing keyword radar, and 1-click bridge into tailored resume rebuilds.
    - **Vernacular Voice AI Mock Interview**: Interactive speech practice in Hindi, Hinglish, and English with STAR framework scoring and real-time confidence metrics.
    - **Skill Gap Roadmap**: 30-day role-specific skill acquisition and capstone project guidance.
    - **Promotion Readiness & Portfolio Analyzer**: Business impact statement conversion and GitHub repository code quality audit.
2. **For Higher Education Institutions & TPOs (B2B — `campus.aivilabs.com`)**:

- **Multi-Tenant Campus Portal**: Isolated tenant branding, custom subdomains, SAML/OIDC SSO integration, and bulk student onboarding wizards.
- **Placement Drive Management**: Shortlisting pipelines, drive scheduling, status updates, and automated candidate communications.
- **Verified Resume Vault & ZIP Exporter**: Streamed zip archive generation of verified PDF resumes for company campus hiring.
- **NAAC & NIRF Accreditation Audit Generator**: Automated generation of official criterion compliance reports and PDF downloads.

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2. System Architecture & High-Level Topology

AIVI utilizes a decoupled, serverless frontend-and-functions architecture paired with a dedicated Hostinger KVM 4 VPS reverse proxy.



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3. Core Technology Stack & Specific Version Registry

Frontend Technologies

Dependency	Version	Purpose
Vite	`^8.2.0`	Next-gen frontend tooling and build engine
React	`^18.3.1`	Core UI component framework
React Router DOM	`^6.30.1`	Declarative client-side routing
TypeScript	`^5.8.3`	Strict type safety and compilation
Tailwind CSS	`^3.4.17`	Utility-first CSS styling engine
Framer Motion	`^12.34.0`	High-performance UI animation library
Firebase SDK	`^12.14.0`	Client Auth, Firestore & Storage integration
Lucide React	`^0.462.0`	Modern SVG iconography
jsPDF	`^4.2.1`	Client-side PDF document compilation
Docx	`^9.6.1`	Word DOCX export generation
Recharts	`^2.15.4`	Radar and analytics chart visualizations
React Hook Form	`^7.61.1`	Form state management
Zod	`^3.25.76`	Schema validation
Vitest	`^4.1.7`	Automated unit testing framework

Backend & Cloud Functions Technologies

Dependency	Version	Purpose
Node.js	`20.x`	Cloud Functions runtime execution engine
Firebase Functions	`^7.2.5`	Serverless backend API endpoint handler
Firebase Admin	`^13.10.0`	Privileged database and auth administration
@google/generative-ai	`^0.24.1`	Official Gemini AI SDK integration
Archiver	`^8.0.0`	Server-side ZIP stream compression
Pdf-Lib	`^1.17.1`	Server-side PDF manipulation & watermark stamping
Nodemailer	`^9.0.3`	SMTP transactional email processing
QRCode	`^1.5.4`	Server-side QR code rendering

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4. Subdomain Routing & Multi-Tenant Segregation Architecture

AIVI enforces strict domain-level segregation to prevent data leakage between public consumer users and institutional campus tenants.

```
// src/utils/domainHelper.ts
export const isCampusDomain = (hostname: string): boolean => {
  return hostname.startsWith("campus.") ||
hostname.includes("campus.staging.aivilabs.com");
};

export const getCampusUrl = (): string => {
  if (typeof window === "undefined") return "https://campus.aivilabs.com";
  return window.location.hostname.includes("staging")
    ? "https://campus.staging.aivilabs.com"
    : "https://campus.aivilabs.com";
};
```

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5. Frontend Application Architecture & Route Registry

All routes are declared in `src/App.tsx` using React lazy loading and suspense boundaries.

Public & Marketing Routes (`aivilabs.com`)

- `/` : B2C Landing Page (`LandingConceptB.tsx`)
- `/auth` , `/login` : User Authentication (`Auth.tsx`)
- `/verify-email` : Email Verification (`VerifyEmail.tsx`)
- `/about` : GEO/LLM-Optimized About Page (`AboutConceptB.tsx`)
- `/how-it-works` : Step-by-Step Workflow Guide (`HowItWorksConceptB.tsx`)
- `/what-is-aivi` : Platform Architectural Overview (`WhatIsAiviConceptB.tsx`)
- `/ai-resume-builder` : Resume Intelligence Product Page (`AIResumeBuilderPageConceptB.tsx`)
- `/job-fit-analysis-tool` : Job Fit Product Page (`JobFitAnalysisToolConceptB.tsx`)
- `/career-gap-analysis-ai` : Skill Roadmap Product Page (`CareerGapAnalysisAIConceptB.tsx`)
- `/interview-intelligence` : Vernacular Voice AI Product Page (`InterviewIntelligenceConceptB.tsx`)
- `/promotion-readiness-ai` : Promotion Readiness Product Page (`PromotionReadinessConceptB.tsx`)
- `/portfolio-analyzer-ai` : Portfolio Showcase Product Page (`PortfolioAnalyzerConceptB.tsx`)
- `/ai-resume-builder-for-freshers` : Fresher Resume Guide (`AIResumeBuilderFreshersConceptB.tsx`)
- `/resume-scoring-software` : ATS Checker Guide (`ResumeScoringSoftwareConceptB.tsx`)
- `/linkedin-optimization` : LinkedIn Optimizer Guide (`LinkedInOptimizationPageConceptB.tsx`)
- `/ai-vs-chatgpt-vs-gemini` : Comparison Guide (`AIVsChatGPTVsGeminiConceptB.tsx`)
- `/contact` : Support & Contact Page (`ContactConceptB.tsx`)
- `/privacy` : Privacy Policy — DPDP Aligned (`Privacy.tsx`)
- `/terms` : Terms of Service (`Terms.tsx`)
- `/dpa` : Data Processing Agreement (`DpaPage.tsx`)

B2C Candidate Dashboard Routes (`/dashboard/*`)

- `/dashboard` : Candidate Dashboard Overview (`DashboardHome.tsx`)
- `/dashboard/builder` : Resume Builder Workspace (`ResumeBuilder.tsx`)
- `/dashboard/resume` : Resume Intelligence Diagnostic (`ResumeIntelligence.tsx`)
- `/dashboard/job-fit` : Job Fit Calculator Tool (`JobFitAnalysis.tsx`)
- `/dashboard/roadmap` : Skill Gap Roadmap (`SkillRoadmap.tsx`)
- `/dashboard/interview` : Vernacular Voice AI Practice (`InterviewPrep.tsx`)
- `/dashboard/promotion` : Promotion Readiness Audit (`PromotionReadiness.tsx`)

- `/dashboard/portfolio`: Portfolio Showcase Audit (`PortfolioAnalyzer.tsx`)
- `/dashboard/linkedin`: LinkedIn Optimizer Workspace (`LinkedInAnalyzer.tsx`)
- `/dashboard/settings`: Account & Entitlements Settings (`SettingsPage.tsx`)

B2B Campus OS Routes (`/tpo/*` & `/campus/*`)

- `/campus/login`, `/campus/login/:tenantCode`: Branded Institutional SSO Login (`BrandedLogin.tsx`)
- `/tpo`: Campus Portal Home (`CampusDashboard.tsx`)
- `/tpo/students`: Bulk Student Directory (`StudentsDirectory.tsx`)
- `/tpo/import`: Student Import Wizard (`StudentImportWizard.tsx`)
- `/tpo/drives`: Placement Drives & Shortlisting Pipeline (`PlacementDrives.tsx`)
- `/tpo/vault`: Verified Student Resume Vault (`ResumeVault.tsx`)
- `/tpo/interviews`: Institutional AI Interview Prep (`CampusInterviews.tsx`)
- `/tpo/accreditation`: NAAC / NIRF Audit Generator (`AccreditationReports.tsx`)
- `/tpo/billing`: Campus Subscription & Usage Metering (`CampusBilling.tsx`)
- `/tpo/calendar`: Campus Placement Calendar (`CampusCalendar.tsx`)
- `/tpo/audit-logs`: Audit Logging Console (`AuditLogsPage.tsx`)
- `/tpo/admin-console`: Platform Superadmin Console (`PlatformAdminConsole.tsx`)

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6. Design System, Styling Tokens & Component Library

AIVI uses Vanilla CSS tokens combined with Tailwind CSS for rich UI aesthetics.

Color Palette Tokens

```css
/* Core Color Palette */
--coral-primary: #FF6D4A;
--coral-hover: #FF7A5B;
--teal-primary: #169C8F;
--teal-accent: #24B7A9;
--slate-dark: #0F172A;
--slate-ink: #112031;
--surface-peach: linear-gradient(180deg, #FFFDF8 0%, #FFF4EE 42%, #F4FAF9 100%);
```

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7. Firebase Integration & Authentication Architecture

Firebase Auth powers user identity across B2C and B2B tenants.

Auth Mechanisms

- **Email/Password**: Standard signup and login flows with email verification (`VerifyEmail.tsx`).
- **Google OAuth**: One-click social sign-in via `signInWithPopup(auth, googleProvider)` .
- **Tenant SSO**: Custom claims (`tenantId` , `role: "tpo" | "student" | "admin"`) injected via Cloud Functions upon login.

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8. Firestore Database Schema & Collection Rules

Security rules enforce multi-tenant isolation (`request.auth.token.tenantId == resource.data.tenantId`).

Major Collections

1. **`users`** : `{ uid, email, displayName, photoURL, createdAt, plan: "free" | "starter" | "pro" }`
2. **`tenants`** : `{ id, slug, name, domain, logoUrl, primaryColor, ssoEnabled, createdAt }`
3. **`students`** : `{ id, tenantId, rollNumber, name, email, department, passoutYear, cgpa, verifiedResumeUrl }`
4. **`placement_drives`** : `{ id, tenantId, companyName, roleTitle, ctc, driveDate, status, shortlistedStudents: [] }`
5. **`resume_vault`** : `{ id, studentId, tenantId, resumeUrl, verifiedAt, verifiedBy, score }`
6. **`accreditation_reports`** : `{ id, tenantId, year, criterion: {}, pdfUrl, generatedAt }`
7. **`ai_usage_logs`** : `{ id, userId, tenantId, feature: "mock_interview" | "resume_rebuild", tokensUsed, costInr, timestamp }`

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9. Cloud Functions Architecture & API Registry

All cloud functions run in `functions/src/index.ts` on Node 20 runtime.

Primary Callable Cloud Functions

1. **`generateAccreditationReport`** : Computes NAAC/NIRF metrics and generates downloadable audit reports.
2. **`exportResumePackage`** : Streams a zip file of verified student PDF resumes for placement drives using `archiver` .
3. **`startMockInterviewTurn`** : Initializes a Sarvam AI / Gemini voice interview turn.
4. **`answerMockInterviewTurn`** : Evaluates candidate spoken answers against STAR criteria.

5. `finishMockInterviewTurn`: Finalizes mock interview turn and records overall scores.
6. `createRazorpayOrder`: Generates Razorpay payment orders.
7. `handleRazorpayWebhook`: Verifies HMAC-SHA256 signatures and provisions plan entitlements.

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10. AI Engine Integration (Gemini, Sarvam AI & Prompt Pipeline)

AIVI pairs deterministic parsing algorithms with generative AI APIs.

AI Pipeline

- **Google Gemini SDK (`@google/generative-ai`)**: Used for structured resume bullet rewriting, JD match keyword extraction, and gap identification.
- **Sarvam AI Engine**: Used for low-latency Speech-to-Text (STT) and Text-to-Speech (TTS) in Hindi, Hinglish, and English during Vernacular Voice AI mock interviews.
- **In-Memory Audio Processing**: Voice audio is processed in live memory streams and is **never stored** on disk or database servers.

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11. Payment Gateway Integration (Razorpay Orders & Webhooks)

Razorpay handles subscription and single-purchase payments.

Payment Execution Flow

1. Client calls `createRazorpayOrder({ planId })` Cloud Function.
2. Backend returns `order_id`, `amount`, `currency`.
3. Client opens Razorpay Checkout modal (`integrate_razorpay_checkout`).
4. Upon payment completion, Razorpay fires `payment.captured` webhook.
5. Cloud Function verifies `x-razorpay-signature` using `crypto.createHmac("sha256", secret)` before granting credits.

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12. Security & Compliance Protocol (DPDP Act 2023 & Security Rules)

AIVI complies with India's **Digital Personal Data Protection (DPDP) Act 2023**.

Compliance Checklist

- **Data Residency**: All candidate data hosted in Google Cloud Mumbai (`asia-south1`).
- **Explicit Consent**: `DpdpConsentModal.tsx` captures affirmative user consent before processing CV documents.
- **Data Minimization**: Voice audio processed strictly in temporary memory; zero audio files stored on servers.

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13. B2B Multi-Tenant Campus OS Architecture

Allows colleges to manage student placement readiness under their own brand.

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14. SAML / OIDC Single Sign-On (SSO) Protocol for Partner Colleges

Partner colleges integrate enterprise identity providers (Microsoft Entra ID, Google Workspace SSO, Shibboleth).

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15. Student Vault & Verified Resume ZIP Exporter Protocol

TPO officers export verified candidate resumes in bulk for company placement drives using `archiver`.

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16. AI Mock Interview & Vernacular Voice AI Mode

Allows candidates to practice interviews in Hindi, Hinglish, or English.

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17. Placement Drives & Candidate Pipeline Management

TPOs track active campus placement drives across 4 pipeline stages.

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18. NAAC & NIRF Accreditation Audit Report Generator

Generates institutional compliance metrics for accreditation bodies (NAAC Criterion 5.2.1 and NIRF Metric 3.a).

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19. Dynamic Legal Shell & Subdomain-Aware Navigation

`LegalHeaderFooterWrapper.tsx` ensures legal subpages (`/dpa` , `/privacy` , `/terms`) render domain-appropriate headers.

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20. Plan Entitlements, Monetization & Billing Architecture

- **Free Tier** (`₹0`): Basic resume audit, 1 Job Fit check.
- **Starter Plan** (`₹399`): Full Resume Studio access, 10 Job Fit checks, 2 AI Mock Interviews.
- **Pro Plan** (`₹599`): Unlimited Resume Studio rebuilds, 50 Job Fit checks, Vernacular Voice AI access.

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21. Environmental Configuration & Secrets Management

Configuration parameters stored in `.env.production` and `.env.staging` .

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22. Infrastructure, Hosting & VPS Deployment Protocols

Frontend bundles are hosted on a Hostinger KVM 4 VPS reverse proxy.

Hostinger VPS Specifications

- **IP Address**: `187.127.134.92`
- **SSH Port**: `22`
- **User**: `root`
- **Password**: `AbhiDev@222020`
- **Nginx Web Root (Staging)**: `/var/www/staging-aivi-campus/html`
- **Nginx Web Root (Production)**: `/var/www/aivi-campus/html`

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23. Deployment Safety Protocol & Staging-First Workflow (Rules D1-D6)

Golden Deployment Rules

- **D1 NEVER DEPLOY AUTOMATICALLY**: Fixing a bug or building a feature does NOT trigger deployment.
- **D2 MCP = READ-ONLY BY DEFAULT**: Server mutations require explicit approval.
- **D3 EVERY CHANGE = COMMIT + PUSH**: Commit with `fix: <what broke>` and push to GitHub.
- **D4 PRODUCTION = PIPELINE ONLY**: Deployment flows through verified scripts.
- **D5 STAGING-FIRST**: Deploy to Staging (`python scripts/deploy_staging.py`) first.
- **D6 PRODUCTION COMMAND**: Deploy to Production ONLY when the user explicitly types `"DEPLOY"` (`python scripts/deploy_production.py`).

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24. Testing & QA Automation (Vitest, Playwright, TestSprite)

- **Unit Testing**: `npm test` runs Vitest test suites.
- **E2E Testing**: Playwright browser automation tests.
- **QA Automation**: TestSprite MCP server integration for frontend test plan execution.

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25. Error Handling, Logging & Observability

- **Cloud Functions Logs**: Viewed via `firebase functions:log` .

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26. Performance Optimization, Caching & Bundle Strategy

- **Vite Rolldown Optimizer**: Production bundle compiled in ~2.5 seconds.

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27. SEO, Social Previews (Open Graph) & Generative Engine Optimization (GEO)

- **Open Graph Image**: `https://aivilabs.com/og-image.png?v=3` (1200x630 high-converting graphic).
- **JSON-LD Schema**: Embedded `Organization` and `SoftwareApplication` schemas for LLM indexers (ChatGPT, Perplexity, Gemini, Claude).

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28. MCP Server Integrations & Autonomous Tooling Registry

Configured in `C:\Users\abhis\.gemini\config\mcp_config.json`:

- `@modelcontextprotocol/server-memory`: Global Knowledge Graph server.
- `@digitalocean/mcp`, `hostinger-api-mcp`, `@playwright/mcp@latest`, `firebase-tools mcp`, `razorpay-mcp-server`.

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29. Administrative Console & Superadmin Management

Superadmin portal accessible at `/tpo/admin-console` (`PlatformAdminConsole.tsx`).

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30. Disaster Recovery, Backup Procedures & Incident Playbook

- **One-Line Rollback Command**: `git checkout main; python scripts/deploy_production.py`

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31. Third-Party Services & Sub-processors Directory

1. **Google Cloud Platform / Firebase** (Auth, Firestore, Cloud Storage, Functions)
2. **Google Gemini AI** (LLM Generative Engine)
3. **Sarvam AI** (Voice STT / TTS Engine)
4. **Razorpay Payments** (INR Payment Gateway)
5. **Hostinger VPS** (Nginx Hosting Infrastructure)

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32. Developer Onboarding, Local Setup & Troubleshooting Guide

Quick Start

```
git clone https://github.com/abhitriloki/aivi-career.git
cd aivi-career
npm install
npm run dev
```

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33. Future Engineering Roadmap & Enhancement Strategy

- **Mobile Native App (React Native)**
- **Automated Resume Parsing from LinkedIn URLs**
- **AI Mock Interview Video Analysis**

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34. Milestone & Fix Log

Recent Milestones & Verified Bug Fixes

- **August 9-13, 2026**:
  - Hero section CTA alignment to ensure 100% above-the-fold visibility.
  - Shortened vertical section gap padding across landing page (`py-8 sm:py-14`).
  - Fixed `.page-offset` top-padding gap (eliminated 98px white space below header on inner pages).
  - Created 6 dedicated public marketing pages under Product menu.
  - Upgraded all 7 AI Tools pages to AIVI v2 features.
  - GEO / LLM-optimized About page with JSON-LD schema.
  - Updated Open Graph image (`/og-image.png?v=3`) for social sharing.
  - Fixed PromoBanner display on home page.
  - Commit Hash: `a18b9be` / `3c226d9`
  - Deployed & Verified on Staging (`https://staging.aivilabs.com`) and Production (`https://aivilabs.com`).